



Actividad Series de Taylor - Series de Maclaurin.

* Ejercicio 1: Serie de Maclaurin de e^x

a) e^{-2x}
- Sustituir

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad x \in \mathbb{R} \Rightarrow e^{-2x} = \sum_{n=0}^{\infty} \frac{(-2x)^n}{n!}$$

$$(ab)^n = a^n b^n$$

$$e^{-2x} = \sum_{n=0}^{\infty} \frac{(-2)^n \cdot x^n}{n!} \quad \text{--- Converge ---} \quad -\infty \quad +\infty$$

Verificar $\sum a_n \Rightarrow \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-2)^{n+1} \cdot x^{n+1}}{(n+1)!}}{\frac{(-2)^n \cdot x^n}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(-2)^{n+1} \cdot x^{n+1} \cdot n!}{(-2)^n \cdot x^n \cdot (n+1)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| -2 \cdot x \cdot \frac{1}{n+1} \right| = |2x| \cdot \lim_{n \rightarrow \infty} \left| \frac{1}{n+1} \right| = |2x| \cdot 0 = 0$$

$R = \infty$ intervalo de convergencia $(-\infty, +\infty) \mid x \in \mathbb{R}$.

b) e^{x^2}

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \quad x \in \mathbb{R}$$

→ Sustituir

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} \quad (-\infty, +\infty) \quad x \in \mathbb{R} \quad \text{converge } +\infty$$

→ Verificar

$$\sum a_n \Rightarrow |a_n| = \left| \frac{x^{2n}}{n!} \right| ; |a_{n+1}| = \left| \frac{x^{2(n+1)}}{(n+1)!} \right| = \left| \frac{x^{2n} \cdot x^2}{(n+1)n!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{2n} \cdot x^2}{(n+1)n!}}{\frac{x^{2n}}{n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{2n} \cdot x^2 \cdot n!}{(n+1) \cdot n! \cdot x^{2n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^2}{n+1} \right| = \frac{x^2}{\infty} = 0$$

$R = \infty$ intervalo de convergencia = $(-\infty, +\infty) \quad x \in \mathbb{R}$

* Ejercicio 2: Serie de Maclaurin de $\cos(x)$

a) $\cos(5x)$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \quad x \in \mathbb{R}$$

(3)

→ Sustituir

$$\cos(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (5x)^{2n} \Rightarrow \cos(5x) = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 5^{2n}}{(2n)!} x^{2n}$$

→ Verificar

$$|a_{n+1}| = \left| \frac{5^{2(n+1)} x^{2(n+1)}}{(2(n+1))!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1 ; |a_n| = \frac{(-1)^n 5^{2n} x^{2n}}{(2n)!} ; |a_{n+1}| = \frac{5^{2n+2} x^{2n+2}}{(2n+2)!}$$

$$\lim_{n \rightarrow \infty} \left| \frac{5^{2n+2} x^{2n+2}}{(2n+2)!} \cdot \frac{(2n)!}{(-1)^n 5^{2n} x^{2n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{5^{2n+2} x^{2n+2} (2n)!}{(2n+2) (-1)^n 5^{2n} x^{2n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{5^2 x^2 (2n)!}{(2n+2)(2n+1)(2n)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{25x^2}{(2n+2)(2n+1)} \right| = \frac{25x^2}{\infty} = 0$$

$R = \infty$ intervalo de convergencia = $(-\infty, +\infty) \quad x \in \mathbb{R}$

b) $\cos(-x^2)$

$$\cos(-x^2) = \cos(x^2)$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} \quad x \in \mathbb{R}$$

→ Sustituir

$$\cos(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (x^2)^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{4n}$$

Verificar $|a_n| = \frac{(-1)^n x^{4n}}{2n!}$; $|a_{n+1}| = \frac{(-1)^{n+1} x^{4n+1}}{(2n+1)!}$; $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{(-1)^{n+1} x^{4n+1}}{(2n+1)!}}{\frac{(-1)^n x^{4n}}{2n!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{4n+1} (2n)!}{(2n+1)! x^{4n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{x^{4n} x^4}{(2n+2)(2n+1)(2n)!} \cdot \frac{(2n)!}{x^{4n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^4}{(2n+2)(2n+1)} \right| = \frac{x^4}{\infty} = 0$$

$R = \infty$ intervalo de convergencia $(-\infty, +\infty)$ $x \in \mathbb{R}$

* Exercício 3: Série de Maclaurin de $\sin(x)$

a) $\sin(3x)$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} \quad x \in \mathbb{R}$$

- Substituir

$$\sin(3x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} (3x)^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n \cdot 3^{2n+1}}{(2n+1)!} x^{2n+1}$$

- Verificar

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1; |a_n| = \left| \frac{3^{2n+1} x^{2n+1}}{(2n+1)!} \right|; a_{n+1} = \left| \frac{3^{2(n+1)+1} x^{2(n+1)+1}}{(2(n+1)+1)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{3^{2(n+1)+1} x^{2(n+1)+1}}{(2(n+1)+1)!} \cdot \frac{(2n+1)!}{3^{2n+1} x^{2n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^{2n+3} x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{3^{2n+1} x^{2n+1}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{3^{2n+3} x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{3^{2n+1} x^{2n+1}} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^2 x^2 (2n+1)!}{(2n+3)(2n+2)(2n+1)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{3^2 x^2 (2n+1)!}{(2n+3)(2n+2)(2n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{3^2 x^2}{(2n+3)(2n+2)} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{9x^2}{(2n+3)(2n+2)} \right| = |9x^2| \lim_{n \rightarrow \infty} \frac{1}{(2n+3)(2n+2)}$$

$$|9x^2| \lim_{n \rightarrow \infty} \frac{1}{\infty} = 9x^2 \cdot 0 = 0$$

$R = \infty$ intervalo de convergencia $(-\infty, +\infty)$ $x \in \mathbb{R}$

b) $\sin(x^2)$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

= Sustituir

$$\sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n (x^2)^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{4n+2}$$

- Verificar

$$|a_{n+1}| = \left| \frac{x^{4(n+1)+2}}{2(n+1)+1!} \right| \quad (6)$$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|; \quad |a_n| = \left| \frac{x^{4n+2}}{(2n+1)!} \right|; \quad |a_{n+1}| = \left| \frac{x^{4n+6}}{(2n+3)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{\frac{x^{4n+6}}{(2n+3)!}}{\frac{x^{4n+2}}{(2n+1)!}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{4n+6} (2n+1)!}{x^{4n+2} (2n+3)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^6 (2n+1)!}{x^2 (2n+3)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{x^4 (2n+1)!}{(2n+3)(2n+2)(2n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^4}{(2n+3)(2n+2)} \right| =$$

$$|x^4| \cdot \lim_{n \rightarrow \infty} \left| \frac{1}{(2n+3)(2n+2)} \right| = |x^4| \cdot \frac{1}{\infty} = 0$$

$R = \infty$ intervalo de convergencia $(-\infty, +\infty)$ $x \in \mathbb{R}$

* Ejercicio 4: Serie de Maclaurin con funciones.

$$a) \frac{1}{1+2x} = \frac{1}{1-(-2x)}$$

Suma geométrica

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

- Sustituir

$$\frac{1}{1-(-2x)} = \sum_{n=0}^{\infty} (-2x)^n$$

$$f(x) = \sum_{n=0}^{\infty} (-1)^n 2^n x^n$$

Verificar $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$; $|a_n| = |-2^n x^n|$ $b^{n+1}/b^n \neq 1$

$$|a_{n+1}| = |-2^{n+1} x^{n+1}|$$

$$\lim_{n \rightarrow \infty} \left| \frac{-2^{n+1} x^{n+1}}{-2^n x^n} \right| = \lim_{n \rightarrow \infty} |-2x| < 1 = |x| < \frac{1}{2}$$

El intervalo válido es $(-\frac{1}{2}; \frac{1}{2})$

b) $f(x) = \frac{1}{1-x^2}$

$$\frac{1}{1-x^2} = \sum_{n=0}^{\infty} (x^2)^n$$

Substituir

$$f(x) = \sum_{n=0}^{\infty} -1 (x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

Verificar $|a_{n+1}| = |x^{2(n+1)}|$

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| \quad |a_n| = |x^{2n}|; \quad |a_{n+1}| = |x^{2n+2}|$$

$$\lim_{n \rightarrow \infty} \left| \frac{x^{2n+2}}{x^{2n}} \right| = \lim_{n \rightarrow \infty} |x^2| = \lim_{n \rightarrow \infty} |x^2| < 1 = x < \sqrt{1}$$

El intervalo válido es $(-1; 1)$

c) $\frac{x+3}{1-x^2} = \frac{(x+3)}{1-(-x^2)} = (x+3)x^2$

$$f(x) = \sum_{n=0}^{\infty} x^n$$

→ Sustituir

$$f(x) = x+3 \sum_{n=0}^{\infty} (-1)^n x^{2n} = x \sum_{n=0}^{\infty} (-1)^n x^{2n} + 3 \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

$$f(x) = \sum_{n=0}^{\infty} (-1)^n x^{2n+1} + 3x^{2n}$$

→ Verificar

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| ; |a_n| = |(x+3)x^{2n}| ; |a_{n+1}| = |(x+3)x^{2(n+1)}|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(x+3)x^{2(n+1)}}{(x+3)x^{2n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{2n+2}}{x^{2n}} \right| = \lim_{n \rightarrow \infty} |x^2|$$

$$\lim_{n \rightarrow \infty} |x^2| < 1 = x < \sqrt{1}$$

El intervalo es válido en $(-1, 1)$

(9)

* Ejercicio 5: Serie de Taylor centrada $x=(x-a)+a$

a) e^{3x} ; $a=-1$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

→ Sustituir

$$e^{3x} = e^{3(x+1)-1} = e^{3(x+1)-3} = e^{-3} \cdot e^{3(x+1)}$$

$$e^{-3} \sum_{n=0}^{\infty} \frac{(3(x+1))^n}{n!} = e^{-3} \sum_{n=0}^{\infty} \frac{3^n (x+1)^n}{n!}$$

→ Verificar

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| ; |a_n| = \left| \frac{e^{-3} 3^n (x+1)^n}{n!} \right| ; |a_{n+1}| = \left| \frac{e^{-3} 3^{n+1} (x+1)^{n+1}}{(n+1)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{e^{-3} 3^{n+1} (x+1)^{n+1} n!}{e^{-3} 3^n (x+1)^n (n+1)!} \right| = \lim_{n \rightarrow \infty} \left| \frac{3(x+1) n!}{(n+1)!} \right| \quad n! = (n+1)!$$

$$\lim_{n \rightarrow \infty} \left| \frac{3(x+1) n!}{(n+1) n!} \right| = \lim_{n \rightarrow \infty} \left| \frac{3(x+1)}{(n+1)} \right|$$

$$3(x+1) \cdot \lim_{n \rightarrow \infty} \frac{1}{n+1} = 3(x+1) \cdot 0 = 0$$

El intervalo válido es $(-\infty; +\infty)$ $x \in \mathbb{R}$

b) $\cos(2x)$; $a = \pi$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n}$$

$x = (x - \pi) + \pi$

coseno de la suma:

$f(x) = \cos(2(x - \pi)) + 2\pi$

$\cos(A)\cos(B) - \sin(A)\sin(B)$

$\cos(2\pi) = 1$; $\sin(2\pi) = 0$

$f(x) = \cos(A)\cos(B) - \sin(A)\sin(B)$ $A = 2(x - \pi)$; $B = 2\pi$

$f(x) = \cos(2(x - \pi))\cos(2\pi) - \sin(2(x - \pi))\sin(2\pi)$

$f(x) = \cos(2(x - \pi)) \cdot 1 - \sin(2(x - \pi)) \cdot 0$

$f(x) = \cos(2(x - \pi))$

- Sustituir

$$\cos(2x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} (2x - \pi)^{2n} = \sum_{n=0}^{\infty} \frac{(-1)^n 2^{2n}}{(2n)!} (x - \pi)^{2n}$$

- Verificar

$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$; $|a_n| = \left| \frac{(-1)^n 2^{2n}}{(2n)!} (x - \pi)^{2n} \right|$

$|a_{n+1}| = \left| \frac{(-1)^{n+1} 2^{2n+1}}{(2n+1)!} (x - \pi)^{2n+1} \right|$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} 2^{2n+1} (x-\pi)^{2n+1}}{(2n+1)!} \cdot \frac{(2n)!}{(-1)^n 2^{2n} (x-\pi)^{2n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{2^{2n+2} (x-\pi)^{2n+2}}{(2n+2)(2n+1)(2n)! 2^{2n} (x-\pi)^{2n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{2^2 (x-\pi)^2}{(2n+2)(2n+1)} \right| = \lim_{n \rightarrow \infty} \left| \frac{4(x-\pi)^2}{\infty} \right| = 0$$

$R = \infty$ el intervalo de convergencia $(-\infty, +\infty)$

c) $\sin(5x)$; $a = -\pi/2$ $x = x - a$

$$x = (x - (-\pi/2)) = x + \pi/2 = (x + \pi/2) - \pi/2$$

$$f(x) = \sin(5(x + \pi/2) - (\pi/2)) \quad \sin(\theta - \phi) = \sin\theta \cos\phi - \cos\theta \sin\phi$$

$$f(x) = \sin\theta \cos\phi - \cos\theta \sin\phi$$

$$\sin(5\pi/2) = 1$$

$$\cos(5\pi/2) = 0$$

$$f(x) = \sin(5(x + \pi/2)) \cdot \cos(5\pi/2) - \cos(5(x + \pi/2)) \sin(5\pi/2)$$

$$f(x) = \sin(5(x + \pi/2)) \cdot 0 - \cos(5(x + \pi/2)) \cdot 1$$

$$f(x) = -\cos(5(x + \pi/2))$$

$$-1(-1)^n = (-1)^{n+1}$$

= Sustituir

$$f(x) = - \sum_{n=0}^{\infty} \frac{(-1)^n (5(x + \pi/2))^{2n}}{(2n)!}$$

- Verificar

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| ; |a_n| = \left| \frac{(-1)^n 5(x + \pi/2)^{2n}}{(2n)!} \right|$$

$$|a_{n+1}| = \left| \frac{(-1)^{n+1} 5(x + \pi/2)^{2n+1}}{2(n+1)!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{(-1)^{n+1} 5(x + \pi/2)^{2n+1}}{2(n+1)!} \cdot \frac{(2n)!}{(-1)^n 5(x + \pi/2)^{2n}} \right| = \lim_{n \rightarrow \infty} \left| \frac{(2n)! 5(x + \pi/2)^{2n+1}}{2(n+1)! 5(x + \pi/2)^{2n}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{5^{2n+2} (x + \pi/2)^2}{5^{2n} (2n+2)(2n+1)2n} \right| = \lim_{n \rightarrow \infty} \left| \frac{25(x + \pi/2)^2}{(2n+2)(2n+1)} \right| = \lim_{n \rightarrow \infty} \frac{25(x + \pi/2)^2}{\infty} = 0$$

$$\lim_{n \rightarrow \infty} \frac{25(x + \pi/2)^2}{\infty} = 0$$

$R = \infty$ el intervalo de convergencia es $(-\infty; +\infty)$ $x \in \mathbb{R}$

d) $\frac{1}{1+7x}$ $a = 21$ $x = (x - 21) + 21$ $\text{radio} = 1$

$$f(x) = \frac{1}{1+7((x-21)+21)}$$

$$\sum_{n=0}^{\infty} t^n \frac{1}{1-t}$$

$$f(x) = \frac{1}{1+7(x-21)+7 \cdot 21} = \frac{1}{1+7(x-21)+147} = \frac{1}{148+7(x-21)}$$

Kut

$$f(x) = \frac{1}{148 \left(1 + \frac{7(x-21)}{148}\right)} = \frac{1}{148} \cdot \frac{1}{1 - \left(-\frac{7}{148}(x-21)\right)}$$

$$x = 1 - \left(-\frac{7}{148}(x-21)\right)$$

Suma de serie geométrica

$$|x| < 1 \Rightarrow \left| -\frac{7}{148}(x-21) \right| < 1 \quad \frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$$

$$|-a| = |a| \quad (-a < x < a)$$

$$\left| \frac{7}{148}(x-21) \right| < 1 \Rightarrow |x-21| < \frac{148}{7} \Rightarrow \frac{-148}{7} < x-21 < \frac{148}{7}$$

$$21 - \frac{148}{7} < x < 21 + \frac{148}{7}$$

- Sustituir $x = -\frac{7}{148}(x-21)$

$$f(x) = \sum_{n=0}^{\infty} (-1)^n \left(\frac{7}{148}\right)^n (x-21)^n = (7x+22)^n = 7^n \left(x + \frac{22}{7}\right)^n$$

El intervalo válido es $\left(21 - \frac{148}{7}; 21 + \frac{148}{7}\right)$

$$e) \frac{9-x}{1+7x} \quad a=21 \quad \begin{array}{l} -x+9 \mid 7x+1 \\ -x-\frac{1}{7} \quad \boxed{\frac{-1}{7}} \end{array} \quad \sum_{n=0}^{\infty} t^n \frac{1}{1-t}$$

$$9-x = \left(-\frac{1}{7}\right)(7x+1) + \frac{16}{7} \quad \boxed{\frac{16}{7}} r$$

$$f(x) = \frac{9-x}{1+7x} = \left(-\frac{1}{7}\right) + \left(\frac{16}{9}\right) \cdot \boxed{\frac{1}{1+7x}} \quad \text{anterior}$$

$$\begin{aligned} 9-x &= (1+7x) \cdot \frac{1}{7} + r \\ 9-x &= \frac{1}{7} + 7x \cdot \frac{1}{7} + r \\ 9 &= \left(\frac{1}{7} + r\right) + (7x) \cdot \frac{1}{7} \\ 9 &= \frac{1}{7} + r \\ -1 &= 7r \end{aligned}$$

$$f(x) = -\frac{1}{7} + \frac{16}{7} \sum_{n=0}^{\infty} (-1)^n \left(\frac{7}{148}\right)^n \cdot (x-21)^n \quad \text{radio} = \frac{148}{7}$$

Separar el 1er término de la serie $-1/7 + 16/7$

$$f(x) = -\frac{1}{7} + \frac{16}{7} \quad a_0 = \frac{15}{7}$$

$$\sum_{n=1}^{\infty} \frac{16}{7} \cdot (-1)^n \cdot \left(\frac{7}{148}\right)^n \cdot (x-21)^n$$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cdot (x-21)^n = \frac{15}{7} + \sum_{n=1}^{\infty} \frac{16}{7} (-1)^n \left(\frac{7}{148}\right)^n$$

El intervalo es válido en $\left(21 - \frac{148}{7}; 21 + \frac{148}{7}\right)$